Code No: 157EM **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
ELECTRIC DRIVES AND CONTROL
(Common to CE, CSE)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Discuss the concept of electric drive and its advantages.
b) Give the classification of electric drives and explain them. [7+8]

2. What are the components of electric drives and discuss the role of each module in detail. [15]

3.a) Explain the multi quadrant operation of a motor driving a load.
b) Draw the closed loop block diagram of electric drive and explain it. [8+7]

4.a) How the moment of inertia of a rotating part is determined in case of electric drive?
b) Explain the concept of load equalization of an electric drive. [8+7]

5.a) Discuss the braking techniques of dc motor and explain them.
b) A 220V, 21A, 1000 rpm dc shunt motor draws armature current of 21 A on full load and runs at1000 rpm. If a resistance of 0.5 ohm is inserted in series with armature. Determine speed at full load torque. [8+7]

6.a) Explain the operation of single phase full converter fed dc motor.
b) Using chopper circuit how the speed of dc motor is controlled. [8+7]

7.a) Explain the speed control of induction motor using stator variable voltage.
b) Discuss the regenerative braking operation of induction motor. [8+7]

8. Write a short note on the following:
a) Losses in electric drive systems
b) Energy efficient operation of drives
c) Maintenance of motors. [5+5+5]

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